SINGAPORE DIMENSIONS • GRADE 6A • CHAPTER 3

Algebraic Expressions & Postfix Traversal

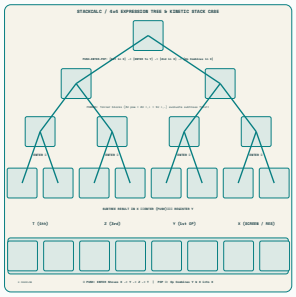
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.6.EE.A.2

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Tree Board

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Build cube towers for variables a=6, b=2, c=8, d=4.

Sketch Model on Paper:
Draw binary expression syntax tree with operator root.

3. Verify with SC-6 (Abstract)

RPN Stack Register Flow:
Postorder traversal: Left child, Right child, Operator. Zero parentheses needed.

Self-Check Keystroke Sequence:

`1` `5` `enter` `5` `+` `8`

`enter` `3` `-` `×`

Expected LCD: 100

1 **Evaluate (12 - 4) × (3 + 2).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Evaluate (25 + 15) ÷ (10 - 2).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Evaluate 8 × (4 + 6) - 20.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Evaluate (50 - 10) ÷ (2 × 4).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 3

Algebraic Expressions & Postfix Traversal • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Interlocking Cubes & 4-Line Hardware LCD Stack

CAD Blueprint: tree_board

Target Device: SC-6 Elementary / SC-32 Bridge

CAD / STL Source File:

Education/build/scad/tree_board.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Charcoal 4x4 Tree Case & Teal Operation Blocks (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Evaluate (12 - 4) × (3 + 2).** Standard: CCSS.MATH.6.EE.A.2	40	[12] [ENTER] [4] [-] [3] [ENTER] [2] [
2	**Evaluate (25 + 15) ÷ (10 - 2).** Standard: CCSS.MATH.6.EE.A.2	5	[25] [ENTER] [15] [+] [10] [ENTER] [2]
3	**Evaluate 8 × (4 + 6) - 20.** Standard: CCSS.MATH.6.EE.A.2	60	[4] [ENTER] [6] [+] [8] [×] [20] [-]
4	**Evaluate (50 - 10) ÷ (2 × 4).** Standard: CCSS.MATH.6.EE.A.2	5	[50] [ENTER] [10] [-] [2] [ENTER] [4]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Highlight that RPN evaluates expressions strictly from the inside out.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.